

Manual

Operation Splitting HLS-AGS 8.2

Version 1.0.23049

Last changed on: 01.09.2020

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1 Overview of Operation Splitting

Purpose

This function package provides functions used to split operations and to schedule split operations in the graphical planning board of HYDRA Shop Floor Scheduling.

Implementation notes

You use the function package if:

- you would like to shorten operation lead times or improve utilization of workplaces / machines available by processing operations simultaneously.

Integration

The function described here is integrated into different function groups in Shop Floor Data Collection. Emphasis here is on data collection, i.e. at the terminal, and the corresponding posting resulting from the collected data.

Features

- Function used to split operations in the graphical planning board in order to plan these split operations simultaneously and to be able to post quantities and times onto these split OPs.
- Splitting up of the total production quantity of an operation into several parallel operations and the option to schedule them at several workplaces / machines .
- Individual definition of the number of splits.
- Automatic indexing for split operations.
- Uploading to the ERP system based on the original operation if an ERP interface is implemented.

2 Splitting Operations

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Purpose

You use this function if you want to shorten the lead time of operations or improve utilization of several workplaces/ machines available for production.

Integration

This function splits an operation into two or more operations. An operation can be split using the following functions:

- [Order sequencing](#).
- [Edit operations](#).
- [Graphic planning](#): can be called up via the context menu of the operation.

Data is always uploaded to the PPS/ ERP system according to the order/ operation number of the original operation.

Requirements

An operation can only be split if the following conditions are met:

- The length of the split number is set to 1 (maximum: 9 splits) or 2 (maximum: 99 splits) in the HYDRA basic parameter settings.
- The identifier "May be split" is set at the operation.
- The operation to be split cannot be a merged operation.
- The operation is not a split operation. It is not allowed to split a split operation once more.
- The operation has not yet started, i.e. it must be marked with status (control indicator) "prepared".

Splitting an operation

Function authorization	op.split
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The function "Split operation"  allows you to split the total production quantity of an operation into several operations. The number of operations subjected to splitting is predefined as per the number of splits configured in the order backlog. If the indicator "May be split" is not set for the operation, you cannot split the operation.

The system runs through the following processing steps when splitting an operation:

Checks

- A check is run to verify whether the operation may be split. If the indicator "May be split" is set for the operation, you can split the operation.
- The number of splits entered must be greater than 1.
- The entered number of splits must be less than the maximum number of splits defined at the operation.
- The operation to be split may not have already been split (split master).
- The operation to be split may itself not be an operation split.
- The operation to be split may not be a merged operation.

If these conditions are met, the operation may be split.

Processing

The system issues the numbers for the splits automatically by distributing a consecutive number. The number is shown in the field "split": The remainder of the number (order/ operation) matches the one of the original operation ("split master"). The number of splits depends on the length/number of characters defined for splits in the HYDRA basic parameter settings.

Example:

Operation 47100100 0100 is divided up into to three splits; the number length for the split defined in the basic settings is 2.

Results of splitting:

Split master: 47100100 0100 (can no longer be logged on)

Individual splits: 47100100 0100 01

47100100 0100 02

47100100 0100 03

When an operation is split, the following values of the original operation are distributed to the splits based on the number of splits:

- Target quantity (primary, secondary, tertiary, base quantity unit)
- Target scrap (primary, secondary, tertiary, base quantity unit)
- Processing time

The quantities are generally divided up in full numbers. If they cannot be distributed evenly, the remainder is allocated to the last operation split.

All other data is transferred from the original operation.

After splitting, the order is flagged for lead time scheduling.



After splitting, only the generated operation splits can be posted. The original operation (split master) is marked as *locked* and cannot be posted. It is no longer shown in the backlog of orders.

The HYDRA Shop Floor Scheduling module no longer shows the split master and, therefore, the master cannot be planned any longer, but only its split operations.

Changes to operation splits, in particular changes in planning (e.g. changes to planned dates or changes to the planned workplace) do not affect the split master.

Please keep in mind that after an operation has been split, any new changes from the ERP system to this original operation ("split master") will have no effect on the individual operations split off.

If the target quantity update is enabled, the operation splits will not be updated, but only the split master.

Cancel operation splits

Function authorization	op.splitrelease
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When an operation split is canceled, all of the operations split off are deleted and only the original operation, which up until now was the split master, continues to exist. The split master is now treated as a normal operation.

Operation splits can only be canceled if all splits of the operation still show a status "prepared".

Displaying operation splits

Whether split operations are displayed depends on the application:

- The original operation (split master) is not shown in graphic planning.
- The split master is shown in the following applications:
 - Edit operations
 - Order information

Whether the single operation splits are displayed also depends on the application:

- The [graphic planning](#) generally shows operation splits. In contrast to this, the split master is no longer shown after splitting.
- The single operations split off are shown in the following applications, provided that the relevant option has been selected:
 - [Edit operations](#)
 - [Order information](#)

Posting to operation splits

Where split operations are concerned, the log record is ONLY generated for the operation split; no log record is generated for the split master itself.

Only the split master's status is updated if the following events occur:

- interrupt operation
- log off operation

Uploading to the PPS/ ERP system

The values recorded or posted to operation splits are generally uploaded to the PPS/ ERP system according to the split master.

Splitting of operations subject to batch management

General

The "split operation" function may also be run for operations subject to management in batches. The function is identical to the split function of conventional operations.

Procedure

The user applies the "split OP" function and splits off an OP from the "split master" in the given operation quantity (primary quantity).

The operation is split. Data relating to material is transferred from the split master to the operations split off.

The required quantity is calculated subject to the input quantity and the quantity split off.

Results after splitting the operation:

With regard to the material, the following data/ values are accepted or used by the operations split off:

- Order header data
 - Finished article (number of the finished article) --> taken over from split master
 - Article name --> taken over from split master
 - Batch number --> taken over from split master (for the batch being generated in this order, e.g. transferred from the ERP; identical for all operation splits)
 - Material type of the finished article --> taken over from split master
- OP data
 - Material number of output material/article --> taken over from split master
 - Material type of output material/article --> taken over from split master
 - Name of output material/article --> taken over from split master
 - Indicator "batch management requirement" --> taken over from split master
 - Indicator "serial numbers required" --> taken over from split master
- Component list
 - All data included in the component list is copied or transferred from the split master.
 - The required quantity is recalculated from the OP split off based on the quantity split off.
- Output batch change:
 - When output batches are changed, the number of the split master is added to the output batch.
- Uploads to the ERP system:
 - Uploads to the ERP system are always performed in relation to the split master.